

When Latent Prediction Is Not Decision Prediction

Decision-Equivalent VJEPA for Action-Relevant World Models

Jupiter Yao
Cornell University
jy2228@cornell.edu

Abstract

World models help agents anticipate consequences before acting, but accurate latent prediction need not preserve distinctions used by a downstream policy. We call this mismatch the *prediction-action gap*. We propose **Decision-Equivalent Variational JEPA (DE-VJEPA)**, which augments variational latent prediction with a KL penalty between the action distributions induced by predicted and true future latents. Evaluation uses behavior-cloned policies whose seeds differ from the policy defining the training loss. Across MetaWorld and PushT, DE-VJEPA lowers independent decision KL relative to VJEPA in seven of eight clean or visually shifted conditions. On clean MetaWorld observations, decision KL falls by 19.8% while latent MSE rises by 15.4%, directly exposing the prediction-action gap; on clean PushT, decision KL falls by 24.0%. Random policy geometry provides no gain, weak behavior cloning provides little gain, and direct action-mean matching does not reproduce the policy-KL result. A Gaussian Product-of-Experts extension improves shifted prediction in some conditions but is not uniformly better. Autoregressive evaluation shows that DE-VJEPA retains lower decision KL through horizon five. However, 14,850 closed-loop MetaWorld episodes across local, progress-aware, and CEM planners do not establish a significant aggregate success improvement over VJEPA, and direct behavior cloning remains strongest on clean inputs. The results therefore support a narrow conclusion: policy-induced geometry improves offline decision preservation, but converting that advantage into control requires a task-grounded planning objective.

1 Introduction

Embodied agents require world models that anticipate the consequences of actions. JEPA-style approaches [LeCun, 2022, Assran et al., 2023, Bardes et al., 2024, Assran et al., 2025] are attractive because they avoid spending capacity on irrelevant high-entropy details by working in latent space. Yet latent-space fidelity and downstream control quality are not the same objective: a predictor can score low reconstruction error while systematically discarding action-critical distinctions, and vice versa.

We call this mis-alignment the *prediction-action gap* and ask whether a policy-induced training signal can reduce it. Our empirical answer is a hybrid objective that

retains latent prediction while adding a *policy-induced equivalence* regularizer, borrowing the conceptual foundations of bisimulation metrics [Ferns et al., 2004].

Contributions.

1. We define and motivate the prediction-action gap in VJEPA world models (§3).
2. We propose hybrid DE-VJEPA, which adds a KL-based decision-equivalence regularizer to variational latent prediction (§4).
3. We characterize the objective through policy-equivalence classes and discuss its collapse modes (§5).
4. We report a leakage-free five-task MetaWorld pilot and three targeted ablations over six source/evaluator-policy pairs (§6).
5. We show that gains require meaningful policy geometry and are not reproduced by random policy KL or direct action-mean matching.
6. We evaluate BJEPAs and DE-BJEPAs as uncertainty-aware Product-of-Experts extensions and identify shift-dependent, rather than universal, gains.
7. We report negative closed-loop results and isolate the remaining integration gap with planner and autoregressive-rollout diagnostics.

2 Background

2.1 Latent World Models

Let o_t denote the observation at time t . An encoder f_ϕ maps observations to latent representations $z_t = f_\phi(o_t)$. A latent predictor g_θ produces future latents from current latents and actions:

$$\hat{z}_{t+k} = g_\theta(z_t, a_{t:t+k}).$$

The standard prediction objective is $\mathcal{L}_{\text{pred}} = d(\hat{z}_{t+k}, z_{t+k})$, where d is Euclidean or cosine distance.

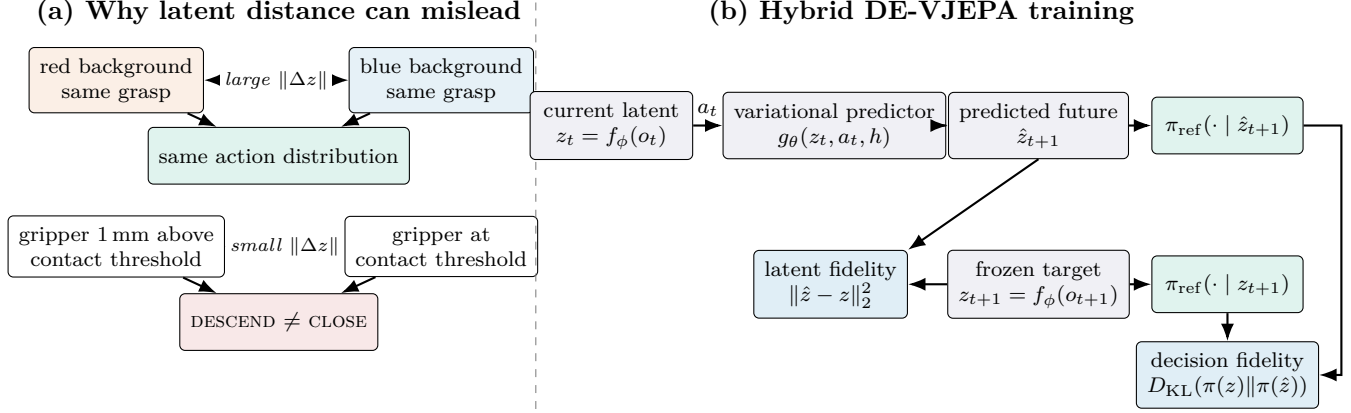


Figure 1: **Prediction-action gap and proposed objective.** (a) Euclidean latent distance can over-penalize decision-irrelevant appearance changes and under-penalize small state changes that cross an action boundary. (b) Hybrid DE-VJEPa retains latent prediction while matching the action distributions induced by predicted and true future latents. The encoder and reference policy are frozen; only the variational predictor is updated.

2.2 Variational JEPA

VJEPa [Huang, 2026] introduces a latent variable h for hidden uncertainty:

$$p_{\theta}(z_{t+k} | z_t, a_{t:t+k}) = \int p_{\theta}(z_{t+k} | z_t, a_{t:t+k}, h) p(h) dh.$$

With variational posterior $q_{\psi}(h | z_t, a_{t:t+k}, z_{t+k})$ the training objective is

$$\mathcal{L}_{\text{VJEPA}} = \mathbb{E}_{q_{\psi}}[-\log p_{\theta}(z_{t+k} | \cdot)] + \beta D_{\text{KL}}(q_{\psi} \| p(h)). \quad (1)$$

2.3 Bisimulation Metrics

Bisimulation [Ferns et al., 2004] defines state equivalence by behavioral similarity: states s, s' are bisimilar if they share the same reward and lead to bisimilar distributions under every action. Deep bisimulation for control [Zhang et al., 2021] learns an encoder ϕ satisfying

$$\|\phi(s) - \phi(s')\| \approx |r(s) - r(s')| + \gamma W(\phi \circ T(\cdot | s), \phi \circ T(\cdot | s')),$$

where W is a Wasserstein distance. Our work is inspired by this but distinct: we apply a *policy-induced* equivalence to the *predictor* objective in an existing JEPA architecture, without re-designing the encoder.

2.4 Related Work and Positioning

Latent predictive representation learning. DE-VJEPa follows the broad JEPA program of learning by predicting in embedding space rather than pixel space [LeCun, 2022, Assran et al., 2023, Bardes et al., 2024]. Recent work extends this idea from masked images to video and action-conditioned world modeling [Garrido et al., 2024, Assran et al., 2025, Destrade et al., 2026]. This

direction is complementary to reconstruction-heavy self-supervised methods such as masked autoencoders [He et al., 2022] and to contrastive predictive learning [Oord et al., 2018]: our question is not only whether a latent is predictable, but whether the predicted latent preserves action-relevant decisions.

World models for control. Classical latent world models learn compact dynamics for planning or policy optimization [Ha and Schmidhuber, 2018, Hafner et al., 2019, 2020, 2021, 2023]. More recent scalable model-based agents such as MuZero, TD-MPC, and TD-MPC2 optimize behavior through latent prediction, value estimation, or model-predictive control rather than direct pixel reconstruction [Schrittwieser et al., 2020, Hansen et al., 2022, 2024]. Generative world-model systems such as UniSim, GAIA-1, Genie, and recent robot world-model surveys emphasize that future prediction is becoming central to embodied AI [Yang et al., 2023, Hu et al., 2023, Bruce et al., 2024, Hou et al., 2026]. DE-VJEPa differs by asking how the *training metric* for a world model should be aligned with downstream decision making.

Robot policies and imitation learning. Our use of a frozen behavior-cloned policy is motivated by the practical success of imitation and visuomotor policy learning, including diffusion policies, robotics transformers, vision-language-action models, and action-chunking transformers [Ross et al., 2011, Chi et al., 2023, Brohan et al., 2022, Zitkovich et al., 2023, Zhao et al., 2023]. At the same time, imitation policies are known to suffer from compounding errors and distribution shift [Ross et al., 2011], which motivates our emphasis on reference-policy quality, inde-

pendent evaluator policies, and uncertainty-aware policy geometry. Cornell-related robot-learning work such as RHyME and X-Sim similarly highlights the importance of action-relevant alignment under mismatched demonstrations or embodiments [Kedia et al., 2024, Dan et al., 2025].

Decision-aware distances and calibration. The theoretical side of DE-VJEPa is closest to bisimulation metrics and invariant representations for control [Ferns et al., 2004, 2011, Zhang et al., 2021]. It is also related to metric learning: rather than assuming Euclidean latent distance is meaningful, we learn or impose a task-dependent geometry [Weinberger and Saul, 2009]. The reference-policy risk is connected to neural-network calibration, where modern networks can output poorly calibrated probabilities even when accurate [Guo et al., 2017]. This motivates the uncertainty-band formulation in Section 4.2.

3 The Prediction–Action Gap

Representation-space distance $\|\hat{z} - z\|^2$ can be miscalibrated for control in two complementary ways.

Remark 1 (Prediction–action gap). *(i) Small latent differences may correspond to different optimal actions (the metric is too lenient). (ii) Large latent differences may be decision-irrelevant (the metric is too strict). A predictor trained only for latent similarity may therefore learn to preserve statistically predictable but decision-irrelevant information, while failing to preserve small action-critical distinctions.*

In embodied manipulation, for instance, a 1mm gripper-cup distance difference may change the optimal action from CLOSE to DESCEND, while a large background-texture change is completely irrelevant. Standard VJEPa treats both equally.

Concrete examples. Consider a button-press trajectory. Changing the table texture can move hundreds of image pixels and therefore shift a generic visual embedding, yet the correct end-effector command is unchanged. Conversely, immediately before contact, two frames may differ by only a few pixels while requiring different commands: continue descending in one state and begin pressing in the other. The first case is a *false positive* for latent distance; the second is a *false negative*. Figure 1 illustrates both cases. The goal of DE-VJEPa is not to discard latent fidelity, but to reweight prediction errors by whether they change the policy distribution.

4 Decision-Equivalent Variational JEPa

4.1 Decision-Equivalence Distance

Let $\pi_{\text{ref}}(a | z, \ell)$ be a frozen reference policy conditioned on latent state z and language instruction ℓ . Instead of measuring whether the predicted latent $\hat{z}$ is close to the target latent z in Euclidean space, we measure whether the two latents induce the same action distribution under π_{ref} :

$$d_{\pi}(\hat{z}, z) = D_{\text{KL}}(\pi_{\text{ref}}(\cdot | z, \ell) \| \pi_{\text{ref}}(\cdot | \hat{z}, \ell)). \quad (2)$$

This distance is asymmetric by design. The forward KL penalizes the predictor when it fails to preserve an action mode that the reference policy assigns high probability to at the true future latent.

The key distinction from standard latent prediction is that d_{π} does not ask whether $\hat{z}$ and z are visually or geometrically identical. It asks whether they are *decision-equivalent* for the current instruction.

4.2 Reliability-Aware Policy Geometry

A practical difficulty is that π_{ref} is usually trained by behavior cloning on expert demonstrations. Such policies can be poorly calibrated outside the demonstration distribution, consistent with broader evidence that modern neural networks often require explicit calibration [Guo et al., 2017]. This matters because the predictor g_{θ} may produce intermediate latents $\hat{z}$ that do not lie exactly on the training-data manifold. In those regions, the gradient from d_{π} may be noisy or even adversarial.

To avoid treating the reference policy as an oracle everywhere, we use a reliability-aware version of the decision distance:

$$d_{\pi, \epsilon}(\hat{z}, z) = [D_{\text{KL}}(\pi_{\text{ref}}(\cdot | z, \ell) \| \pi_{\text{ref}}(\cdot | \hat{z}, \ell)) - \epsilon(\hat{z}, z)]_+, \quad (3)$$

where $[x]_+ = \max(x, 0)$. The tolerance $\epsilon(\hat{z}, z)$ forms a small uncertainty band around the reference policy’s action distribution. When the policy is confident and the predicted latent lies near the training support, ϵ is small and the loss behaves like the original KL. When the predicted latent is far from the trusted region of π_{ref} , the band prevents the model from overfitting to unreliable policy gradients.

One simple instantiation is

$$\epsilon(\hat{z}, z) = \epsilon_0 + c(u(\hat{z}) + u(z)), \quad (4)$$

where $u(z)$ is an uncertainty score. In experiments, $u(z)$ can be estimated using policy entropy, ensemble disagreement, or latent distance to the demonstration manifold. This modification does not add a new training objective; it only prevents the decision-equivalence loss from trusting π_{ref} in regions where the policy geometry is unreliable.

4.3 DE-VJEPa Objectives

The pure decision-equivalence objective is

$$\begin{aligned} \mathcal{L}_{\text{DE-VJEPa}} = & \mathbb{E}_{q_\psi} [d_{\pi, \epsilon}(\hat{z}_{t+k}, z_{t+k})] \\ & + \beta D_{\text{KL}}(q_\psi(h \mid z_t, a_{t:t+k}, z_{t+k}) \parallel p(h)). \end{aligned} \quad (5)$$

Here h is the latent stochastic variable from VJEPa, and

$$\hat{z}_{t+k} \sim p_\theta(\cdot \mid z_t, a_{t:t+k}, h).$$

The encoder f_ϕ and the reference policy π_{ref} are frozen. Only the predictor g_θ and the variational posterior are updated.

Compared with standard VJEPa, the only structural change is the prediction criterion. Instead of training the model to reproduce the exact target latent, DE-VJEPa trains it to preserve the target latent’s action-relevant content.

Empirically, replacing latent prediction entirely is unnecessarily destructive to representation fidelity. We therefore evaluate the hybrid objective

$$\mathcal{L}_{\text{hybrid}} = \mathcal{L}_{\text{VJEPa}} + \lambda D_{\text{KL}}(\pi_{\text{ref}}(\cdot \mid z_{t+k}) \parallel \pi_{\text{ref}}(\cdot \mid \hat{z}_{t+k})). \quad (6)$$

The latent term anchors predictions to the learned representation manifold, while the policy term emphasizes errors that alter the reference policy distribution. The pure objective in Eq. (5) remains a theoretical limiting case.

4.4 Multi-Step Decision Equivalence

A single-step policy KL captures immediate action consistency, but world models are ultimately useful because they support multi-step reasoning. To connect DE-VJEPa more directly to planning, we also define a multi-step extension:

$$d_\pi^{(K)} = \sum_{j=1}^K \gamma^{j-1} D_{\text{KL}}(\pi_{\text{ref}}(\cdot \mid z_{t+j}, \ell) \parallel \pi_{\text{ref}}(\cdot \mid \hat{z}_{t+j}, \ell)). \quad (7)$$

This objective asks the predictor to preserve the sequence of future decisions, not merely the next decision. The single-step version is still meaningful as a first stage: if a downstream planner re-queries π_{ref} or another policy at every imagined rollout step, then local decision consistency at each predicted state is already useful. However, for long-horizon planning, Eq. (7) is the more direct objective.

A value-based alternative is

$$d_V(\hat{z}, z) = |V^{\pi_{\text{ref}}}(\hat{z}, \ell) - V^{\pi_{\text{ref}}}(z, \ell)|, \quad (8)$$

which compares predicted and true latents by their estimated long-run value. We treat d_V as an important ablation rather than the default objective, because accurate

value estimation is harder than estimating a local policy distribution.

Relationship to bisimulation. DE-VJEPa is inspired by bisimulation but does not claim to implement full bisimulation. Classical bisimulation compares reward and transition structure. DE-VJEPa instead compares the action distributions induced by a frozen policy. Thus, it is better understood as a policy-induced behavioral equivalence relation: two latents are equivalent when they lead the agent to make the same decision.

5 Theoretical Motivation

5.1 Policy-Equivalence Classes

The correct theoretical object in DE-VJEPa is not injectivity of π_{ref} . In practice, many different latents should induce the same action distribution. For example, two visually different backgrounds may require exactly the same robot action. Therefore, DE-VJEPa should not be interpreted as recovering the true latent state. It should be interpreted as recovering the true latent state up to *policy equivalence*.

Definition 1 (Policy equivalence). *For a fixed instruction ℓ , define*

$$z \sim_\pi z' \iff \pi_{\text{ref}}(\cdot \mid z, \ell) = \pi_{\text{ref}}(\cdot \mid z', \ell).$$

The equivalence class of z is

$$[z]_\pi = \{z' : z' \sim_\pi z\}.$$

Proposition 1 (Collapse characterization). *For any two latents z and $\hat{z}$,*

$$d_\pi(\hat{z}, z) = 0 \iff \hat{z} \in [z]_\pi.$$

Therefore, DE-VJEPa collapses exactly those latent distinctions that are invisible to the reference policy.

Proof. By the non-negativity of KL divergence, $D_{\text{KL}}(p \parallel q) = 0$ iff $p = q$ almost everywhere. Applying this to

$$p = \pi_{\text{ref}}(\cdot \mid z, \ell), \quad q = \pi_{\text{ref}}(\cdot \mid \hat{z}, \ell),$$

we obtain

$$d_\pi(\hat{z}, z) = 0 \iff \pi_{\text{ref}}(\cdot \mid z, \ell) = \pi_{\text{ref}}(\cdot \mid \hat{z}, \ell).$$

By the definition of $\sim_\pi$, this is equivalent to $\hat{z} \in [z]_\pi$. $\square$

Corollary 1 (Injective policy as a special case). *If $\pi_{\text{ref}}(\cdot \mid z, \ell)$ is injective in z over the support of the predictor, then*

$$d_\pi(\hat{z}, z) = 0 \iff \hat{z} = z.$$

Thus, exact latent recovery is a special case of DE-VJEPa, not the general goal.

Remark 2 (Why non-injectivity is desirable). *A highly expressive policy will not be globally injective. In low-entropy regions, many latents may map to the same high-confidence action. This is not a failure of DE-VJEPa. It is the intended behavior: the model should not waste predictive capacity on state differences that do not change the decision. The risk is not non-injectivity itself, but wrong non-injectivity, where π_{ref} collapses states that a better policy would distinguish. This is why reference-policy quality and uncertainty calibration are central empirical questions.*

5.2 When Can DE-VJEPa Fail?

The collapse characterization also identifies the main failure mode. If the reference policy ignores a task-critical feature, then DE-VJEPa will also ignore that feature. Formally, if there exist two latents z and z' such that

$$\pi_{\text{ref}}(\cdot | z, \ell) = \pi_{\text{ref}}(\cdot | z', \ell)$$

but the optimal policy satisfies

$$\pi^*(\cdot | z, \ell) \neq \pi^*(\cdot | z', \ell),$$

then DE-VJEPa will incorrectly treat z and z' as equivalent. This makes the quality of π_{ref} not merely an implementation detail, but a core assumption of the method.

6 Experiments

6.1 Setup

We use 33,016 frames from 500 successful demonstrations across five MetaWorld tasks [Yu et al., 2020]: REACH, BUTTON-PRESS, DRAWER-OPEN, DOOR-OPEN, and PICK-PLACE. The visual encoder is a frozen ImageNet ResNet-18 proxy, not an official V-JEPa backbone. We train Gaussian behavior-cloned policies and variational one-step latent predictors with seeds 42, 1000, and 10000.

Leakage-free evaluation. The true future latent is never provided to the predictor at evaluation time. Each trained predictor is evaluated by policies trained with seeds different from its training reference policy, producing six source/evaluator-policy pairs. We report raw latent MSE, independent forward decision KL, and independent action-mean MSE on clean observations and deterministic brightness, blur, and desaturation shifts. Statistical tests pair methods by source/evaluator pair.

6.2 Pilot Reproduction

At $\lambda = 0.01$, clean latent MSE increases by 15.4% ($p = 2.2 \times 10^{-18}$), while independent decision KL and action MSE decrease by 19.8% ($p = 6.5 \times 10^{-14}$) and

Table 1: VJEPa/BJEPa comparison. Values are mean $\pm$ standard error across 30 task/source/evaluator units. Lower is better.

Shift	Method	Latent MSE	Decision KL	Action MSE
Clean	VJEPa	.00151 $\pm$.00007	.350 $\pm$.024	.000734 $\pm$.000130
	BJEPa	.00205 $\pm$.00014	.426 $\pm$.023	.001000 $\pm$.000159
	DE-VJEPa	.00174 $\pm$.00007	.280 $\pm$.021	.000668 $\pm$.000112
	DE-BJEPa	.00205 $\pm$.00015	.361 $\pm$.021	.000932 $\pm$.000149
Brightness	VJEPa	.05121 $\pm$.00059	308.5 $\pm$ 66.4	.07941 $\pm$.00573
	BJEPa	.04878 $\pm$.00074	215.5 $\pm$ 41.9	.07248 $\pm$.00534
	DE-VJEPa	.05113 $\pm$.00062	235.8 $\pm$ 52.8	.07436 $\pm$.00546
	DE-BJEPa	.04731 $\pm$.00073	206.2 $\pm$ 38.4	.07089 $\pm$.00512

9.0% ($p = 0.0028$). Under brightness, decision KL decreases by 23.6% ($p = 0.0023$) with no latent-MSE change ($p = 0.608$). Figure 2 summarizes the complete pattern.

6.3 Lambda Trade-off

We sweep λ over six values on three tasks. The best clean compromise is 10^{-3} : decision KL and action MSE fall by 12.6% and 6.0%, while latent MSE rises by 0.9%. Larger weights eventually degrade all clean metrics (Figure 4a).

6.4 Does Meaningful Policy Geometry Matter?

Random policy KL is indistinguishable from baseline; weak BC changes decision KL by only -0.26% clean and -3.81% under brightness. Strong BC produces the large gains (Figure 3).

6.5 Policy KL Versus Action-Mean Matching

We compare policy KL with the simpler regularizer $\|\mu_{\text{ref}}(z) - \mu_{\text{ref}}(\hat{z})\|^2$, sweeping its weight from 10^{-4} to 10^{-1} . Even at its strongest weight, mean matching reduces clean decision KL by only 3.3% and is ineffective under brightness; policy KL reduces the same metrics by 21.1% and 24.4% (Figure 4b).

6.6 Bayesian Product-of-Experts Extension

We additionally compare VJEPa and DE-VJEPa with BJEPa and DE-BJEPa. BJEPa combines a learned diagonal-Gaussian dynamics expert with an empirical task-conditioned Gaussian prior using a Product of Experts (PoE). The prior is fitted only on training latents and never observes the held-out future latent.

The Bayesian extension is shift-dependent. DE-VJEPa is best on clean decision KL; DE-BJEPa is best on all three brightness metrics; BJEPa is best on blur decision KL. None improves desaturated decision KL. Thus uncertainty modeling is complementary, not uniformly superior.

6.7 Cross-Dataset Baseline Study

We next run a matched 11-condition comparison on both the five-task MetaWorld archive and LeRobot PushT.

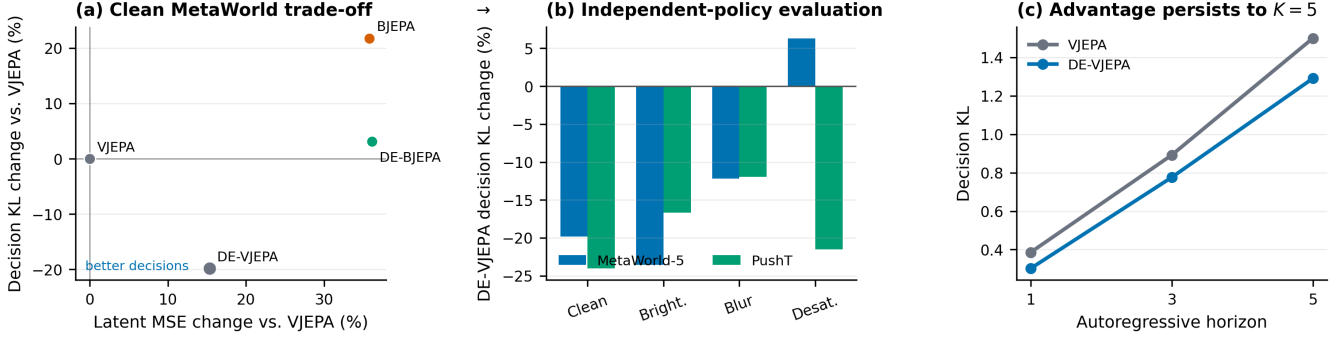


Figure 2: **Offline prediction-action results.** (a) On clean MetaWorld observations, DE-VJEPA moves into the lower-right quadrant: decision KL improves even though latent MSE worsens. BJEPa shows the opposite trade-off. (b) Relative to matched VJEPA predictors, DE-VJEPA lowers independent decision KL in seven of eight dataset-condition comparisons; MetaWorld desaturation is the exception. Percentages are computed from the paired DE-VJEPA/VJEPA means. (c) In held-out autoregressive rollouts, the decision-KL advantage remains visible through horizon five.

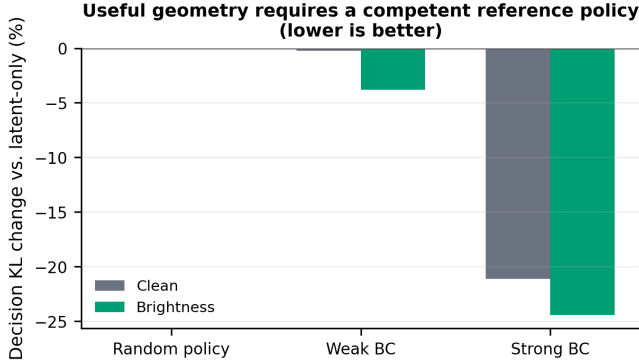


Figure 3: **Reference-policy quality ablation.** Random geometry produces no measurable benefit, weak behavior cloning produces only a small shift, and strong behavior cloning produces the decision-KL reduction. This negative control argues against interpreting the auxiliary KL as generic regularization.

In addition to VJEPA, BJEPa, DE-VJEPA, and DE-BJEPa, the matrix includes persistence, linear action-conditioned dynamics, a no-action predictor, cosine latent prediction, direct action-mean matching, random-policy KL, and weak-BC-policy KL. PushT contains 25,650 frames from 206 episodes at 10 Hz. Its two-dimensional actions are absolute pixel targets, which we map from the $[0, 512]$ workspace to $[-1, 1]$ before BC training.

Decision KL improves in seven of eight cells. All four PushT paired tests are significant ($p \leq 0.014$), but PushT has only six source/evaluator pairs. Persistence remains best on clean PushT, so the defensible claim is improvement over trained VJEPA rather than universal domi-

Table 2: DE-VJEPA change relative to VJEPA in the matched cross-dataset study. Negative values are improvements.

Dataset	Shift	Latent MSE	Decision KL	Action MSE
MetaWorld-5	Clean	+15.4%	−19.8%	−9.0%
	Brightness	−0.2%	−23.6%	−6.4%
	Blur	+1.1%	−12.2%	−0.9%
	Desaturation	+1.4%	+6.3%	+0.8%
PushT	Clean	−3.5%	−24.0%	−13.7%
	Brightness	+1.1%	−16.7%	−8.4%
	Blur	−1.6%	−11.9%	−5.1%
	Desaturation	+0.6%	−21.5%	−12.0%

nance.

6.8 Extended Baseline Matrix

Figure 5 replaces a winner-take-all summary with the full effect-size matrix. It shows both DE-VJEPA’s consistency and the important exceptions: persistence on clean PushT, cosine prediction under PushT desaturation, and Bayesian variants under MetaWorld brightness or blur.

6.9 Closed-Loop MetaWorld Evaluation

We evaluate 1,620 episodes over three tasks, three predictor seeds, three visual conditions, and 20 matched episodes per cell. A seed-7 BC policy, independent of predictor training, either acts directly or scores 32 one-step candidate actions using predicted policy entropy. Full planner details are in Appendix B.

DE-VJEPA exceeds VJEPA by 1.30 success points overall ($p = 0.473$, 95% CI $[-2.41, 4.44]$) and 2.78 points under brightness ($p = 0.179$); neither is significant. BC-only remains 22.22 points stronger on clean observations. Across a further 13,230 episodes, no local or

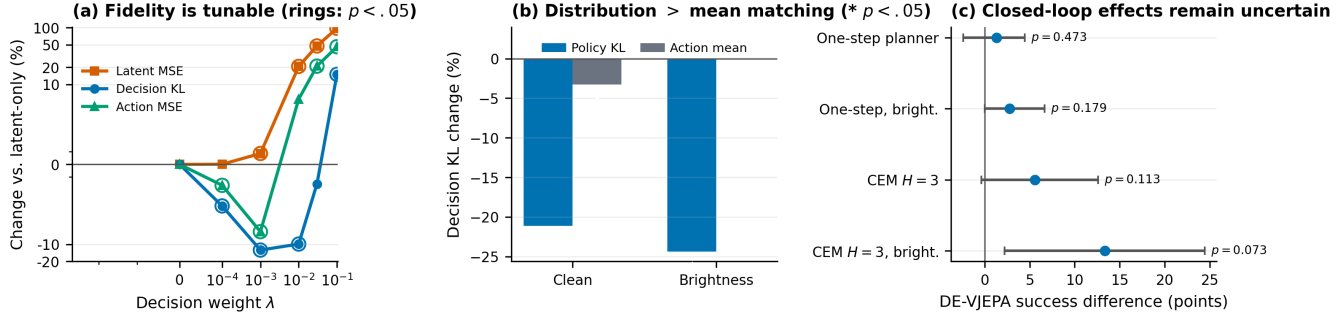


Figure 4: **Ablation and inferential-statistics dashboard.** (a) Clean-data effect sizes across the decision-loss weight; rings mark paired $p < 0.05$. (b) Full-distribution KL is substantially stronger than matching only the Gaussian policy mean. (c) Closed-loop success differences with paired bootstrap 95% confidence intervals. Positive point estimates do not exclude zero, although the horizon-three brightness result is suggestive.

progress-aware planner establishes a significant aggregate advantage. The strongest directional result is CEM at $H = 3$ under brightness: +13.33 points ($p = 0.073$, CI [2.22, 24.44]). Figure 4c displays the principal confidence intervals.

7 Discussion

What the experiments establish. The central empirical result is metric decoupling: a predictor can become worse under Euclidean latent error while becoming better under independently evaluated policy behavior. The effect appears across two datasets and persists in autoregressive rollouts. The strong/weak/random reference-policy ordering further shows that the gain depends on meaningful action geometry.

What they do not establish. Lower policy KL is not itself a control objective. The closed-loop experiments show that local entropy, action-consistency, and trajectory-progress scores do not reliably turn more decision-faithful predictions into better actions. In particular, planners that deviate from the BC mean can lose more from action selection error than they gain from improved prediction. DE-VJEPa should therefore be viewed as a world-model objective whose control value depends on a compatible value or planning objective, not as a complete model-based controller.

Practical implication. Latent MSE alone is insufficient for selecting world models intended for control. At minimum, evaluation should report both representation error and a held-out decision metric, preferably using policies not used to define the training loss. When the two disagree, downstream planning experiments must de-

termine whether the decision metric captures useful task structure or merely local policy imitation.

8 Limitations

Reference policy quality. d_π is only as good as π_{ref} . A poor reference policy may define a decision geometry that ignores task-critical features.

Bias inheritance. If π_{ref} ignores certain observations, the world model will too—even if a better policy would attend to them.

KL direction. We use the forward KL; the reverse KL would penalize mode-seeking. Neither is universally correct; the choice warrants empirical validation.

Continuous action spaces. Computing d_π exactly requires evaluating KL divergences between continuous distributions, which is simple for Gaussian policy heads but more difficult for flow- or diffusion-based policies [Chi et al., 2023].

Short-horizon limitation. Decision equivalence defined via a single-step policy may not capture long-run value. A natural extension replaces d_π with $d_V(\hat{z}, z) = |V^\pi(\hat{z}) - V^\pi(z)|$.

Empirical scope. The offline demonstrations are all successful, so demonstration-level success correlations are not identifiable, and the ResNet-18 encoder is a practical proxy rather than a large pretrained video backbone. MetaWorld contains 30 paired task-training-seed-evaluator-seed units per offline visual condition,

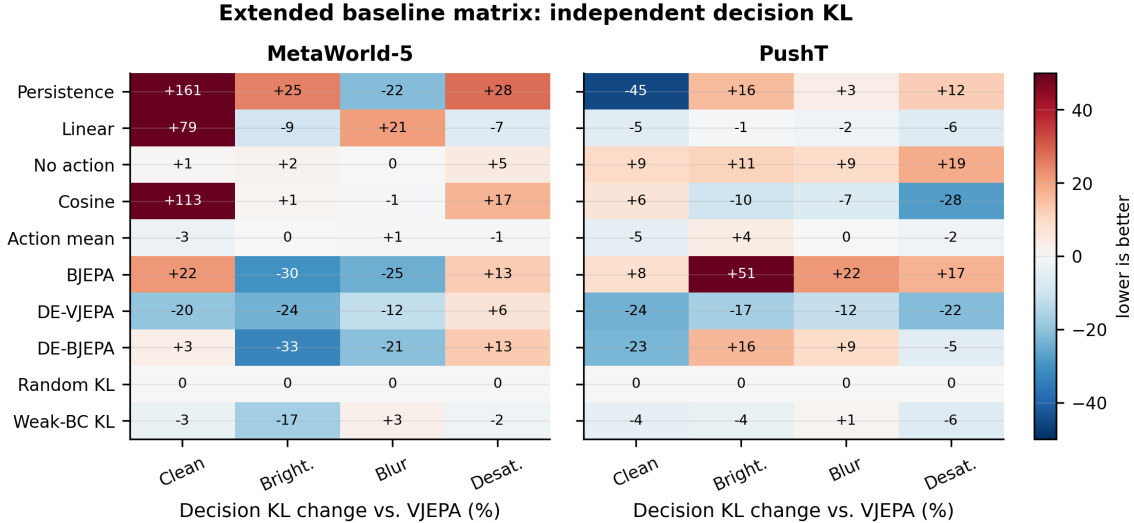


Figure 5: **Ten-method decision-KL matrix.** Each cell is the percentage change from VJEPA; blue is better. DE-VJEPA is the only learned nonlinear predictor with improvements in seven of eight dataset-condition cells. Persistence and cosine prediction win isolated PushT cells, while BJEPA variants are strongest for selected MetaWorld shifts. Values are clipped to $\pm 50\%$ for color scaling but printed without clipping.

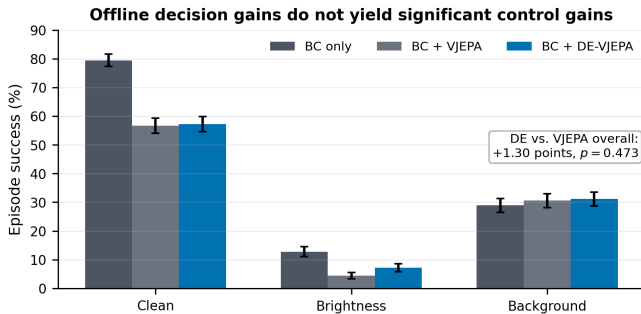


Figure 6: **Closed-loop transfer.** Direct BC is strongest on clean observations. DE-VJEPA is directionally better than VJEPA in aggregate, but the paired difference is not significant. Error bars are binomial standard errors over 180 episodes per method and condition.

whereas the single-task PushT study contains only six paired training/evaluator combinations. DE-VJEPA improves offline decision KL over VJEPA in seven of eight dataset-condition comparisons, but MetaWorld desaturation is negative and persistence remains stronger on clean PushT.

The closed-loop studies contain 14,850 episodes in total: the original 1,620-episode transfer study, a 4,860-episode local-score ablation, a 4,860-episode one-step progress-planner study, and a 3,510-episode CEM/MPC study. They cover only three tasks and one independently seeded BC controller. The learned score is normalized time

within successful demonstrations, not reward-to-go or a true value function. The background shift is a deterministic image transformation rather than a physically randomized simulator texture. Neither the local-score planners nor the progress-aware planners establish a statistically significant aggregate DE-VJEPA advantage over matched VJEPA, and longer CEM horizons often degrade clean success. These restrictions preclude claims of general visual robustness, successful model-based control, or universal superiority.

9 Conclusion

We proposed hybrid DE-VJEPA, which supplements variational latent prediction with a policy-induced KL regularizer. It reduces independent decision KL in seven of eight MetaWorld/PushT conditions, sometimes while latent MSE worsens, and the effect requires a competent reference policy. The advantage persists through five-step autoregressive prediction.

The closed-loop evidence is deliberately negative: none of the tested planners produces a significant aggregate success gain. DE-VJEPA therefore establishes better *decision preservation*, not yet better control. The remaining challenge is to pair this geometry with a task-grounded value objective and a planning horizon matched to model training.

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A Reproducibility Appendix

A.1 Dataset and Evaluation Units

The archive contains 33,016 encoded frames from 500 successful MetaWorld demonstrations, with 100 episodes for each of REACH, BUTTON-PRESS, DRAWER-OPEN, DOOR-OPEN, and PICK-PLACE. Removing transitions that cross episode boundaries gives 32,516 valid one-step prediction examples before splitting. Observations are rendered at 224×224 from the `corner2` camera and encoded once by a frozen ImageNet ResNet-18 into 512-dimensional latents. Actions are four-dimensional continuous MetaWorld controls.

For each seed in $\{42, 1000, 10000\}$, episodes are shuffled and split 80/20 into training and validation sets. Splitting is performed by episode rather than by frame. Each trained predictor is evaluated by the two behavior-cloned evaluator policies whose seeds differ from its training/reference-policy seed. Consequently, each visual condition contains 5 tasks $\times$ 3 training seeds $\times$ 2 evaluator seeds = 30 paired experimental units. Frame-level metrics are saved, but hypothesis tests first average frames within each task/source/evaluator unit to avoid treating correlated frames as independent replicates.

PushT. The second dataset is the official LeRobot PushT release (<https://huggingface.co/datasets/lerobot/pusht>). It contains 25,650 frames from 206 single-task episodes, with 96×96 RGB observations, two-dimensional agent state, and two-dimensional target-position actions at 10 Hz. We decode the provided AV1 video, retain the published episode indices, and encode each frame with the same frozen ResNet-18 used for MetaWorld. PushT actions are pixel coordinates with observed range approximately $[12, 511]$; they are transformed as $a' = a/256 - 1$ so the bounded Gaussian policy mean can represent the action domain. An initial unnormalized run caused policy collapse and is explicitly archived as invalid; none of its metrics is used in the paper.

PushT uses the same three training seeds and two non-matching evaluator seeds per source seed. Because it has one task, each condition has six paired source/evaluator units rather than MetaWorld’s 30 task/source/evaluator units.

A.2 Visual Conditions

Clean evaluation uses the frozen archive without image modification. The three held-out image transformations are deterministic: brightness is multiplied by 0.55, Gaussian blur uses radius 2.0, and color saturation is multiplied by 0.15. The same frame ordering, episode identifiers, actions, and task labels are retained in every shifted

archive, allowing sample-aligned evaluation.

A.3 Model and Optimization Details

Latents and actions are standardized using statistics fitted only on the training partition for each seed. The same normalization is stored with each BC checkpoint and reused by its corresponding predictor. Predictor validation and final evaluation use deterministic predictive means; stochastic samples are used only for the variational training path.

A.4 Product-of-Experts BJEPa

The dynamics expert predicts diagonal Gaussian parameters $(\mu_{\text{dyn}}, \sigma_{\text{dyn}}^2)$ from the current latent, action, task embedding, and variational state. The prior expert is an empirical diagonal Gaussian $(\mu_{\text{prior}, \ell}, \sigma_{\text{prior}, \ell}^2)$ fitted for task ℓ on training target latents. It does not use z_{t+1} at inference. The combined prediction is

$$\tau_B = \sigma_{\text{dyn}}^{-2} + \sigma_{\text{prior}}^{-2}, \quad (9)$$

$$\sigma_B^2 = \tau_B^{-1}, \quad (10)$$

$$\mu_B = \sigma_B^2 \left(\sigma_{\text{dyn}}^{-2} \mu_{\text{dyn}} + \sigma_{\text{prior}}^{-2} \mu_{\text{prior}} \right). \quad (11)$$

All operations are elementwise. Log variances are clipped to $[-10, 5]$, variances are lower-bounded by 10^{-6} before inversion, and training aborts on non-finite precision, variance, mean, or loss.

A.5 Training Algorithm

A.6 Complete Four-Method Results

A.7 Paired Statistical Comparisons

Against VJEPa, DE-VJEPa’s clean decision-KL change is -0.0693 (-19.8% ; paired t -test $p = 6.5 \times 10^{-14}$; Wilcoxon $p = 1.9 \times 10^{-9}$), while clean latent MSE changes by $+0.000231$ ($+15.4\%$; $p = 2.2 \times 10^{-18}$). Under brightness, DE-BJEPa versus VJEPa changes latent MSE by -0.00390 (-7.6% , $p = 6.6 \times 10^{-12}$), decision KL by -102.3 (-33.2% , $p = 0.0044$), and action MSE by -0.00852 (-10.7% , $p = 0.00093$). Under blur, BJEPa versus VJEPa reduces decision KL by 150.6 (24.5%, $p = 0.0073$). Under desaturation, BJEPa increases decision KL by 421.3 (12.6%, $p = 0.024$), demonstrating that the extension is not uniformly robust.

DE-BJEPa improves clean decision KL relative to BJEPa by 0.0651 ($p = 4.2 \times 10^{-11}$), but remains worse than DE-VJEPa on clean decision KL by 0.0803 ($p = 4.8 \times 10^{-11}$). Under brightness, DE-BJEPa has lower means than DE-VJEPa for all three metrics, although the decision-KL difference does not reach 0.05 significance ($p = 0.089$). These comparisons motivate the narrow conclusion that Bayesian predictive structure helps under brightness and blur, while policy-KL regularization is most reliable on clean observations.

Table 3: Concrete implementation and training configuration used in the reported four-method comparison.

Component	Setting	Details
Encoder	Frozen ResNet-18	ImageNet weights; 512-D latent; no fine-tuning
Reference/evaluator policy	Gaussian BC	2-layer MLP, width 512, SiLU, 32-D task embedding
Policy output	Diagonal Gaussian	Tanh mean; log standard deviation clipped to $[-5, 1]$
Policy training	AdamW	30 epochs, batch 1024, lr 3×10^{-4} , weight decay 10^{-5}
Predictor context	z_t, a_t, ℓ	One-step horizon; 32-D task embedding
Variational state	Gaussian h	64 dimensions; log variance clipped to $[-10, 5]$
Predictor network	Residual MLP	2 hidden layers, width 1024, LayerNorm, SiLU
Predictor training	AdamW	50 epochs, batch 512, lr 3×10^{-4} , weight decay 10^{-5}
Regularization	Gradient clipping	Global norm clipped to 5.0; AMP enabled
VAE weight	$\beta = 10^{-3}$	KL from posterior $q(h z_t, a_t, z_{t+1})$ to standard normal
Decision weight	$\lambda_{DE} = 10^{-2}$	Used by DE-VJEPa and DE-BJEPa
BJEPa prior	Empirical task Gaussian	Per-task mean/variance from normalized training latents only
BJEPa objective	Gaussian NLL	Diagonal predictive NLL plus variational KL
Hardware	RTX 4090	Single 24 GB GPU; eight data-loader workers

A.8 Metrics, Calibration, and Saved Artifacts

For each held-out transition we save $\|z_{t+1} - \hat{z}_{t+1}\|_2^2$, $D_{KL}(\pi_{\text{eval}}(\cdot | z_{t+1}) \| \pi_{\text{eval}}(\cdot | \hat{z}_{t+1}))$, and $\|\mu_{\text{eval}}(z_{t+1}) - \mu_{\text{eval}}(\hat{z}_{t+1})\|_2^2$. BJEPa variants additionally save diagonal Gaussian NLL and mean predictive variance. Calibration is assessed by quantile-binning predictions according to variance and plotting bin-level variance against latent error and decision KL.

The raw core file is `results/vjepa_bjepa_de_comparison.csv`; aggregate estimates are in `results/vjepa_bjepa_de_comparison_summary.csv`; prespecified paired tests are in `results/vjepa_bjepa_de_comparison_paired.csv`. The associated table and plots are generated by:

```
python scripts/run_experiment.py \
--config configs/core_vjepa_bjepa_de.yaml
python scripts/analyze_results.py \
--config configs/core_vjepa_bjepa_de.yaml
```

A.9 Extended Baseline Matrix

The matched two-dataset study uses the following conditions:

1. **Persistence:** $\hat{z}_{t+1} = z_t$, with no trainable predictor.
2. **Linear dynamics:** a single residual linear layer conditioned on z_t, a_t , and task identity.
3. **No-action VJEPa:** the standard predictor with a_t replaced by zero, testing whether action conditioning contributes.
4. **Cosine JEPa:** latent cosine distance replaces latent MSE.
5. **VJEPa:** the latent-MSE variational predictor.

6. **Action-MSE:** VJEPa plus direct policy-mean matching at weight 0.1.
7. **Random-policy KL:** VJEPa plus KL from a frozen random policy.
8. **Weak-policy KL:** VJEPa plus KL from a two-epoch BC policy.
9. **DE-VJEPa:** VJEPa plus strong-BC policy KL at weight 0.01.
10. **BJEPa:** Gaussian PoE prediction trained with latent NLL.
11. **DE-BJEPa:** BJEPa plus strong-BC policy KL at weight 0.01.

The raw and aggregate outputs are `results/cross_method_metaworld.csv`, `results/cross_method_metaworld_summary.csv`, `results/cross_method_pusht.csv`, and `results/cross_method_pusht_summary.csv`. Paired comparisons are stored in the corresponding `_paired.csv` files. The invalid pre-normalization PushT run is retained only for auditability under an `_invalid_action_scale` filename.

B Closed-Loop Rollout Research Record

B.1 Candidate-Action Planning Algorithm

Why this planner is diagnostic. The planner does not use reward, privileged state, or a learned value function. It tests whether a predictor that better preserves policy geometry makes local BC action selection more stable. Its failure to beat BC-only does not imply that the learned dynamics are useless for every planner; it shows that local entropy minimization is insufficient in the present setting.

Algorithm 1: Leakage-free training and evaluation of VJEPA/BJEPA variants

1. Split demonstration episodes into seed-specific training and validation partitions; fit latent/action normalization on training data only.
2. Train a Gaussian BC reference policy on normalized training latents. For each source seed, reserve the policies from the other two seeds as evaluators.
3. For BJEPA variants, fit the task-conditioned empirical Gaussian prior using normalized training latents only.
4. For each minibatch, encode context (z_t, a_t, ℓ) and infer $q_\psi(h \mid z_t, a_t, z_{t+1}, \ell)$ for the variational training path.
5. Predict $\hat{z}_{t+1}$. VJEPA uses a residual point predictor; BJEPA combines dynamics and task-prior experts with the diagonal Gaussian PoE.
6. Compute the base loss: latent MSE for VJEPA or predictive Gaussian NLL for BJEPA. Add $\beta D_{\text{KL}}(q_\psi(h) \parallel \mathcal{N}(0, I))$.
7. For DE variants, add $\lambda_{\text{DE}} D_{\text{KL}}(\pi_{\text{ref}}(\cdot \mid z_{t+1}) \parallel \pi_{\text{ref}}(\cdot \mid \hat{z}_{t+1}))$.
8. Update predictor parameters with AdamW. The encoder and reference policy remain frozen.
9. At evaluation, remove the true future latent from predictor inputs, use the predictive mean, and score with independently seeded evaluator policies.

B.2 Closed-Loop Configuration**B.3 Task-Level Closed-Loop Results**

The aggregate pattern is heterogeneous. DE-VJEPA improves over VJEPA on clean reach and button press, and under brightness for both tasks, but it reduces clean drawer-open success. No method solves shifted drawer open. Background shift produces the only aggregate condition where predictor-assisted methods slightly exceed BC-only, but the improvement is driven by reach and is not statistically significant.

B.4 Closed-Loop Statistical Tests

The prespecified paired unit is the task–predictor–seed–visual-condition success rate, yielding 27 units overall and nine per condition. DE-VJEPA versus VJEPA gives:

- overall success difference +1.30 points, standard error 1.78, paired t -test $p = 0.473$, bootstrap CI $[-2.41, 4.44]$;
- clean +0.56 points, $p = 0.915$, CI $[-10.00, 8.89]$;
- brightness +2.78 points, $p = 0.179$, CI $[0.00, 6.67]$;
- background shift +0.56 points, $p = 0.681$, CI $[-1.67, 2.78]$;
- overall return difference +10.14, $p = 0.363$, CI $[-10.24, 32.66]$.

DE-VJEPA versus BC-only gives an overall success difference of -8.52 points ($p = 0.076$, CI $[-17.96, -0.74]$),

clean difference -22.22 points ($p = 0.081$), brightness difference -5.56 points ($p = 0.095$), and background difference $+2.22$ points ($p = 0.719$). The bootstrap intervals operate on the 27 paired aggregate units; episode-bootstrap intervals in Figure 6 shows uncertainty in each raw success proportion. These intervals answer a different question from the paired-unit bootstrap and should not be interchanged.

B.5 Online Prediction Diagnostics

For the executed candidate, the evaluator encodes the realized next frame and computes online prediction metrics. Averaged over 180 episodes per condition, DE-VJEPA has lower latent MSE than VJEPA under clean (0.00498 versus 0.00560), brightness (0.00720 versus 0.00802), and background shift (0.01127 versus 0.01265). It also has lower action MSE in all three conditions. Decision KL is lower under clean (2.34 versus 3.46) and background shift (46.0 versus 55.9), but similar and very large under brightness (52.1 versus 51.4). Thus, the online diagnostics largely reproduce the offline ordering without producing a significant task-success gain.

B.6 Raw Rollout Data Dictionary

Each row of `results/closed_loop_rollout_raw.csv` is one completed episode. The columns are:

method	BC-only or predictor-assisted controller.
task	MetaWorld environment name.
seed	Predictor-training seed; repeated for BC-only to preserve paired cells.

Table 4: Complete core comparison. Entries are mean $\pm$ standard error across 30 paired task/source/evaluator units. Lower is better. Bold denotes the lowest mean within each shift and metric.

Shift	Method	Latent MSE	Decision KL	Action MSE
Clean	VJEPa	0.001506 $\pm$ 0.000067	0.3496 $\pm$ 0.0237	0.000734 $\pm$ 0.000130
	BJEPa	0.002045 $\pm$ 0.000145	0.4257 $\pm$ 0.0230	0.001000 $\pm$ 0.000159
	DE-VJEPa	0.001738 $\pm$ 0.000074	0.2803 $\pm$ 0.0207	0.000668 $\pm$ 0.000112
	DE-BJEPa	0.002051 $\pm$ 0.000149	0.3606 $\pm$ 0.0207	0.000932 $\pm$ 0.000149
Brightness	VJEPa	0.05121 $\pm$ 0.00059	308.5 $\pm$ 66.4	0.07941 $\pm$ 0.00573
	BJEPa	0.04878 $\pm$ 0.00074	215.5 $\pm$ 41.9	0.07248 $\pm$ 0.00534
	DE-VJEPa	0.05113 $\pm$ 0.00062	235.8 $\pm$ 52.8	0.07436 $\pm$ 0.00546
	DE-BJEPa	0.04731 $\pm$ 0.00073	206.2 $\pm$ 38.4	0.07089 $\pm$ 0.00512
Blur	VJEPa	0.2413 $\pm$ 0.0043	613.7 $\pm$ 95.0	0.1458 $\pm$ 0.0122
	BJEPa	0.2218 $\pm$ 0.0043	463.1 $\pm$ 58.2	0.1355 $\pm$ 0.0120
	DE-VJEPa	0.2440 $\pm$ 0.0045	539.0 $\pm$ 75.5	0.1445 $\pm$ 0.0120
	DE-BJEPa	0.2130 $\pm$ 0.0048	486.4 $\pm$ 63.8	0.1344 $\pm$ 0.0119
Desaturation	VJEPa	0.1983 $\pm$ 0.0046	3335 $\pm$ 1081	0.3290 $\pm$ 0.0546
	BJEPa	0.1964 $\pm$ 0.0044	3756 $\pm$ 1193	0.3344 $\pm$ 0.0553
	DE-VJEPa	0.2010 $\pm$ 0.0047	3546 $\pm$ 1135	0.3315 $\pm$ 0.0559
	DE-BJEPa	0.1915 $\pm$ 0.0043	3766 $\pm$ 1198	0.3308 $\pm$ 0.0542

policy_seed	Independent deployment-policy seed, fixed to 7.	success diagnostic.
visual_condition	Clean, brightness, or background shift.	mean_action_norm Mean Euclidean norm of executed actions.
episode_id	Episode index within a task-seed-condition cell.	mean_policy_entropy Mean BC Gaussian entropy.
environment_seed	Deterministic reset seed shared by methods.	mean_prediction_latent_mse Mean online next-latent error.
planner_type	BC action or one-step candidate planner.	mean_decision_kl Mean online independent policy KL.
num_candidate_actions	Candidate count, 32 for this study.	mean_action_mse Mean online policy-mean error.
lambda_de	Requested decision-regularizer checkpoint weight.	C Supplementary Integration-Gap Diagnostics
success	Binary MetaWorld success indicator.	
return	Sum of environment rewards.	C.1 Planner Ablation Protocol The planner ablation preserves the tasks, predictor seeds, independent seed-7 BC controller, visual conditions, episode horizon, candidate count, and action noise of the original closed-loop evaluation. It changes only the candidate score. Candidate perturbations are regenerated deterministically from the matched environment seed and step index, so VJEPa and DE-VJEPa receive
episode_length	Executed environment steps.	
final_distance_to_goal	Final simulator-provided object-to-target distance when available.	
failure_reason	Success, timeout, termination, or no-	

Algorithm 2: One-step closed-loop evaluation with an independent BC controller

1. **Inputs:** frozen encoder f , independent seed-7 BC policy π_{BC} , optional predictor g_θ , task ℓ , visual condition c , candidate count $N = 32$, noise scale $\sigma_a = 0.12$, and deviation weight $\alpha = 0.1$.
2. Reset MetaWorld with the prespecified environment seed. Render the RGB frame, apply condition c , and encode $z_t = f(o_t)$.
3. Query the BC distribution $p_t = \pi_{\text{BC}}(\cdot | z_t, \ell)$ and set $a_t^{(1)} = \mu(p_t)$.
4. For **bc_only**, execute $a_t^{(1)}$ and skip to Step 8.
5. For a predictor-assisted method, sample $a_t^{(i)} = \text{clip}(a_t^{(1)} + \epsilon_i, -1, 1)$ for $i = 2, \dots, N$, where $\epsilon_i \sim \mathcal{N}(0, \sigma_a^2 I)$.
6. Predict one-step latents $\hat{z}_{t+1}^{(i)} = g_\theta(z_t, a_t^{(i)}, \ell)$ using the predictive mean.
7. Score each candidate by

$$S_i = H\left[\pi_{\text{BC}}(\cdot | \hat{z}_{t+1}^{(i)}, \ell)\right] + \alpha \|a_t^{(i)} - a_t^{(1)}\|_2^2,$$
 and execute $a_t^{(i^*)}$ for $i^* = \arg \min_i S_i$.
8. Observe reward, termination, success, and the next rendered frame. Encode the realized z_{t+1} and, for world-model methods, log online latent MSE, independent decision KL, and action-mean MSE between z_{t+1} and the selected prediction.
9. Repeat until success, environment termination, or 200 steps. Append the completed episode to the raw CSV immediately so interrupted runs can resume.

Figure 7: Closed-loop candidate-action planning and leakage-free evaluation.

exactly the same candidate actions. The evaluated scores are:

$$S_{\text{ent}}(a_i) = H[\pi_{\text{BC}}(\cdot | \hat{z}_{t+1}^{(i)})] + \alpha \|a_i - a_{\text{BC}}\|_2^2, \quad (12)$$

$$S_{\text{KL}}(a_i) = D_{\text{KL}}\left(\pi_{\text{BC}}(\cdot | z_t) \parallel \pi_{\text{BC}}(\cdot | \hat{z}_{t+1}^{(i)})\right) + \alpha \|a_i - a_{\text{BC}}\|_2^2, \quad (13)$$

$$S_\mu(a_i) = \|\mu_{\text{BC}}(\hat{z}_{t+1}^{(i)}) - a_{\text{BC}}\|_2^2 + \alpha \|a_i - a_{\text{BC}}\|_2^2. \quad (14)$$

A random-candidate control samples the same candidate set but selects an index uniformly. BC-only is evaluated once per matched cell. This produces 4,860 episodes:

540 BC episodes + $4 \times 2 \times 540$ predictor-score episodes.

C.2 Planner Ablation Statistical Results

The paired unit is task-predictor-seed-visual-condition, yielding 27 units overall and nine within each condition. DE-VJEPa versus VJEPa gives:

- entropy score: +1.11 success points, $p = 0.680$, bootstrap CI $[-4.07, 6.30]$;
- decision-consistency score: -1.11 points, $p = 0.597$, CI $[-5.19, 2.78]$;
- action-mean consistency: exactly 0.00 points;
- random-candidate control: exactly 0.00 points because candidate generation and random selection are paired and do not depend on the predictor.

No planner score establishes a DE-VJEPa success advantage over VJEPa.

Relative to BC-only, entropy scoring reduces VJEPa success by 9.07 points ($p = 0.0215$) and DE-VJEPa success by 7.96 points ($p = 0.0564$). Decision-consistency scoring reduces VJEPa success by 5.37 points ($p = 0.0341$) and DE-VJEPa success by 6.48 points ($p = 0.00737$). Random candidate selection reduces success by 2.22 points ($p = 0.0202$). Action-mean consistency has zero success difference from BC-only.

Action deviation explains the ordering. The mean root-mean-square action deviation from BC is approximately 0.12–0.15 for entropy and decision-consistency scoring, 0.107 for the random control, and less than 6×10^{-4} for action-mean consistency. Thus, the only planner that matches BC success is also the planner that effectively declines to use the world model. The experiment does not show that the predictors contain no control-relevant information; it shows that the tested local scores do not convert that information into better actions.

Condition-specific behavior. Entropy-scored DE-VJEPa is directionally better than entropy-scored VJEPa under brightness (6.67% versus 4.44%) and background shift (32.22% versus 31.11%), but neither difference is significant. Decision consistency performs poorly

Table 5: Complete closed-loop evaluation configuration.

Field	Value
Tasks	REACH-V3, BUTTON-PRESS-V3, DRAWER-OPEN-V3
Methods	BC-only, BC+VJEPA, BC+DE-VJEPA
Predictor seeds	42, 1000, 10000
Deployment/evaluation policy seed	7; distinct from predictor seeds
Visual conditions	clean, brightness factor 0.55, deterministic background tint
Episodes	20 per task-seed-condition-method cell
Total episodes	1,620
Maximum horizon	200 environment steps
Candidate planner	32 candidates; Gaussian action noise 0.12
Candidate score	predicted BC entropy + 0.1 action-deviation penalty
DE regularizer checkpoint	$\lambda_{DE} = 0.01$
Encoder	frozen ImageNet ResNet-18, 512-dimensional latent
Action selection	deterministic policy and predictor means
Primary endpoint	episode success
Secondary endpoints	return, length, final goal distance, action norm, policy entropy, online latent MSE, decision KL
Compute	NVIDIA RTX 4090; approximately 3.3 wall-clock hours including resume

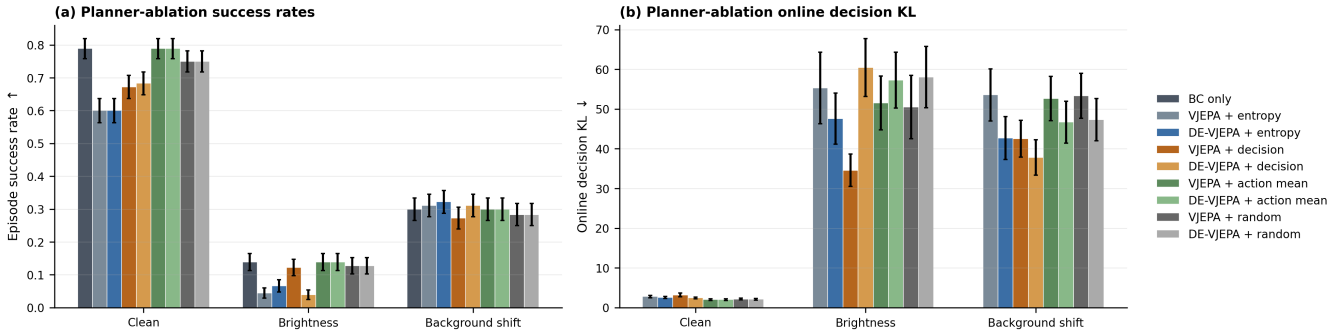


Figure 8: Planner-ablation success (left) and online decision KL (right). Improved online decision metrics do not imply improved task success. The action-mean score remains almost exactly at the BC action and reproduces the BC-only success rate.

for DE-VJEPA under brightness (3.89%) despite improved predicted decision geometry elsewhere. This is evidence that minimizing local distribution change can favor conservative or dynamically unhelpful actions.

C.3 Autoregressive Multi-Step Diagnostic

To separate planner failure from rapid model compounding error, trained one-step VJEPA and DE-VJEPA predictors are rolled autoregressively over held-out MetaWorld demonstration sequences. The predictor receives the recorded action sequence but never receives future latents. Horizons $K \in \{1, 3, 5\}$ are scored with the independent seed-7 BC evaluator. The final raw file contains 63,846 method-sample-horizon rows.

DE-VJEPA reduces decision KL relative to VJEPA by approximately 20.5% at $K = 1$, 12.8% at $K = 3$, and 13.7% at $K = 5$. Its mean decision-error slope is also

lower (0.255 versus 0.288 per prediction step). In contrast, DE-VJEPA latent MSE is higher at every horizon, with latent-error slopes 0.001085 versus 0.000958. The decision advantage therefore weakens from its one-step maximum but does not disappear by $K = 5$.

C.4 Revised Diagnosis of the Integration Gap

The combined evidence rejects a simple explanation in which DE-VJEPA’s decision advantage immediately collapses under autoregressive prediction. Instead:

1. DE-VJEPA preserves lower decision KL through five prediction steps.
2. DE-VJEPA does not significantly beat VJEPA under any tested local candidate score.
3. Entropy and decision-consistency scores underperform direct BC.

Table 6: Task-level closed-loop success rates (%). Each cell contains 60 episodes: 20 episodes for each of three predictor seeds.

Task	Method	Clean	Brightness	Background shift
Reach	BC only	40.0	11.7	8.3
	BC + VJEPA	31.7	6.7	20.0
	BC + DE-VJEPA	35.0	10.0	16.7
Button press	BC only	98.3	26.7	78.3
	BC + VJEPA	85.0	6.7	71.7
	BC + DE-VJEPA	95.0	11.7	76.7
Drawer open	BC only	100.0	0.0	0.0
	BC + VJEPA	53.3	0.0	0.0
	BC + DE-VJEPA	41.7	0.0	0.0

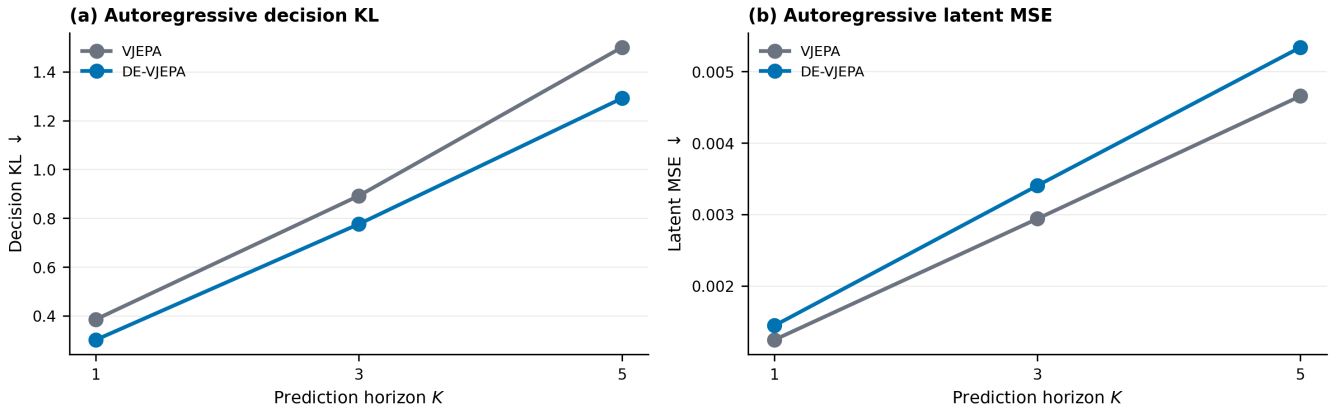


Figure 9: Autoregressive decision KL (left) and latent MSE (right). DE-VJEPA maintains lower decision KL through horizon five while retaining the expected latent-fidelity penalty.

- Random perturbations also hurt, showing that candidate exploration itself has a measurable cost.
- Action-mean consistency matches BC only by producing negligible action deviations, not by exploiting predicted futures.

Merely replacing one local consistency score with another is unlikely to close the control gap.

D Supplementary Progress-Aware Planning Results

D.1 Progress-Head Validation

The supported diagnosis is therefore: *offline and multi-step decision-preserving predictions are measurable, but local policy-consistency objectives are not task-value objectives*. A useful next planner must score long-horizon progress, reward, success probability, or value while controlling model uncertainty. Appropriate next experiments are CEM/MPC over predicted latents, a learned value head, multi-step DE training aligned to the planning horizon, and policy improvement on imagined trajectories.

The available MetaWorld demonstrations contain successful trajectories but do not store reward-to-go or a task-specific distance target. We therefore train a task-conditioned *progress head*, not a value function, using $y_t = t/(T - 1)$. The MLP has two 512-unit hidden layers and is trained independently with seeds 42, 1000, and 10000. Table 9 reports validation statistics averaged over

Table 7: Planner-ablation success rates (%). Each method–score–condition cell contains 180 episodes.

Method	Planner score	Clean	Brightness	Background	Overall
BC only	Direct BC mean	78.89	13.89	30.00	40.93
VJEPa	Entropy	60.00	4.44	31.11	31.85
DE-VJEPa	Entropy	60.00	6.67	32.22	32.96
VJEPa	Decision consistency	67.22	12.22	27.22	35.56
DE-VJEPa	Decision consistency	68.33	3.89	31.11	34.44
VJEPa	Action-mean consistency	78.89	13.89	30.00	40.93
DE-VJEPa	Action-mean consistency	78.89	13.89	30.00	40.93
VJEPa	Random candidate	75.00	12.78	28.33	38.70
DE-VJEPa	Random candidate	75.00	12.78	28.33	38.70

Table 8: Multi-step diagnostic averaged across reach, button press, and drawer open. Lower is better.

Method	K	Latent MSE	Decision KL	Action MSE
VJEPa	1	0.001336	0.3778	0.000289
DE-VJEPa	1	0.001545	0.3004	0.000267
VJEPa	3	0.003233	0.9050	0.000688
DE-VJEPa	3	0.003727	0.7888	0.000694
VJEPa	5	0.005170	1.5311	0.001124
DE-VJEPa	5	0.005886	1.3214	0.001138

Table 9: Validation of the 512-unit trajectory-progress head. Correlation is with normalized time within a successful demonstration.

Task	MSE ↓	Pearson ↑	Spearman ↑
All tasks	0.000356	0.99794	0.99796
Reach	0.000990	0.99437	0.99431
Button press	0.000200	0.99897	0.99911
Drawer open	0.000078	0.99956	0.99956

those seeds.

The head easily reconstructs trajectory position. This validates the implementation and provides a smooth planning score, but it does not establish that the score measures causal task progress. In particular, a planner can increase predicted trajectory time while moving to an off-demonstration latent.

D.2 One-Step Progress Planning

The one-step study evaluates 4,860 episodes over three tasks, three predictor seeds, three visual conditions, four action-deviation weights, and matched VJEPa/DE-VJEPa predictors, with a shared BC-only control. At

Table 10: Progress-aware CEM success rates (%) at $\alpha = 0.1$. Each method–horizon–condition cell contains 90 episodes.

Method	H	Clean	Brightness	Background	Overall
BC only	–	80.00	12.22	27.78	40.00
VJEPa+CEM	1	76.67	13.33	30.00	40.00
DE-VJEPa+CEM	1	81.11	12.22	28.89	40.74
VJEPa+CEM	3	24.44	13.33	27.78	21.85
DE-VJEPa+CEM	3	26.67	26.67	28.89	27.41
VJEPa+CEM	5	18.89	15.56	25.56	20.00
DE-VJEPa+CEM	5	15.56	17.78	26.67	20.00

$\alpha = 0.1$, where actions remain close to BC, VJEPa and DE-VJEPa obtain 40.56% and 40.93% aggregate success, respectively. Their condition-level success rates are 77.78/78.89% on clean observations, 13.33/13.33% under brightness, and 30.56/30.56% under background shift. The paired aggregate difference is +0.37 percentage points ($p = 0.161$).

Smaller penalties make the planner more active but less reliable. At $\alpha = 0.01$, DE-VJEPa improves over VJEPa under brightness by 9.44 points (24.44% versus 15.00%), but the paired result is not significant ($p = 0.231$), and clean success falls to 41.11%. At $\alpha = 0.3$, both predictor planners nearly reproduce BC. The one-step result therefore does not establish successful model-based control.

D.3 Progress-Aware CEM/MPC

CEM uses population 128, three refinement iterations, horizons $H \in \{1, 3, 5\}$, and action penalties $\alpha \in \{0.01, 0.1\}$. The complete study contains 3,510 episodes. Table 10 reports the more stable $\alpha = 0.1$ setting.

At $H = 1$, the two model-based planners are close to BC overall, and the DE-VJEPa–VJEPa difference is

+0.74 points ($p = 0.537$). At $H = 3$, DE-VJEPa exceeds VJEPa by 5.56 points overall ($p = 0.113$, paired 95% bootstrap CI $[-0.37, 12.59]$) and by 13.33 points under brightness ($p = 0.073$, CI $[2.22, 24.44]$). The effect is directionally consistent with the offline decision-preservation claim but does not cross the prespecified significance threshold. At $H = 5$, aggregate success is identical (20.00%).

D.4 Result and Remaining Test

The progress-aware experiments refine, but do not overturn, the paper’s negative closed-loop conclusion. They show a potentially meaningful DE-VJEPa advantage under horizon-three planning and brightness shift, while also showing substantial clean-control degradation as the planning horizon grows. The current evidence is compatible with two explanations: DE-VJEPa provides a better latent rollout for moderate-horizon planning, or the observed difference is sampling variation around a weak progress objective. The next decisive experiment is horizon-matched $K = 3$ predictor training followed by the same $H = 3$ CEM evaluation. Stronger-backbone evaluation is separately required before attributing the offline result to representation-independent decision geometry.

E Auxiliary Scaffolding Artifacts

The release archive ships two scaffolding subsystems that were prototyped during the closed-loop investigation but are deliberately *not* part of the load-bearing experimental evidence in this paper. We document them here so the artifact inventory is self-explanatory and so follow-up work can pick them up without rediscovering the wiring.

Value-head planner (`value_planner_*`). A scalar state-value head trained on top of frozen VJEPa / DE-VJEPa features, intended to provide a learned terminal cost for CEM rollouts (configs `value_head*.yaml` and `value_planner*.yaml`; runner `train_value_head.py`; summaries and figures named `value_planner*.{csv,png}`). On MetaWorld pilot tasks the head converges and calibrates reasonably (smoke calibration plot), but combining the head with CEM did not produce a closed-loop improvement that survived seed averaging, so this branch was superseded by the progress-aware planner reported in Appendix D (`cem_mpc_progress512` artifacts). The full sweep CSVs and plots are retained as a negative-result record and as a starting point for value-aware planning experiments that share the same DE-VJEPa backbone.

`smoke_dinov2.py`). A backbone-substitution harness that swaps the VJEPa encoder for a frozen DINOv2 ViT and reruns the offline decision-KL pipeline. The harness exists to support the “stronger-backbone evaluation is separately required” caveat noted above, but the corresponding training sweep has not been launched in this submission. Only the launch scripts, configs, and smoke test are shipped; no DINOv2 results table or figure is included, and none of the main-text claims depend on this branch. We retain the files so the substitution experiment is reproducible end-to-end as soon as compute is available.

Both subsystems are isolated: removing the `value_planner*/value_head*` and `*dinov2*` files would not change any numeric result, table, or figure reported in Sections 6–8 of the main text.

DINOv2	backbone	validation
(<code>backbone_validation_dinov2,</code>		<code>pilot_dinov2,</code>

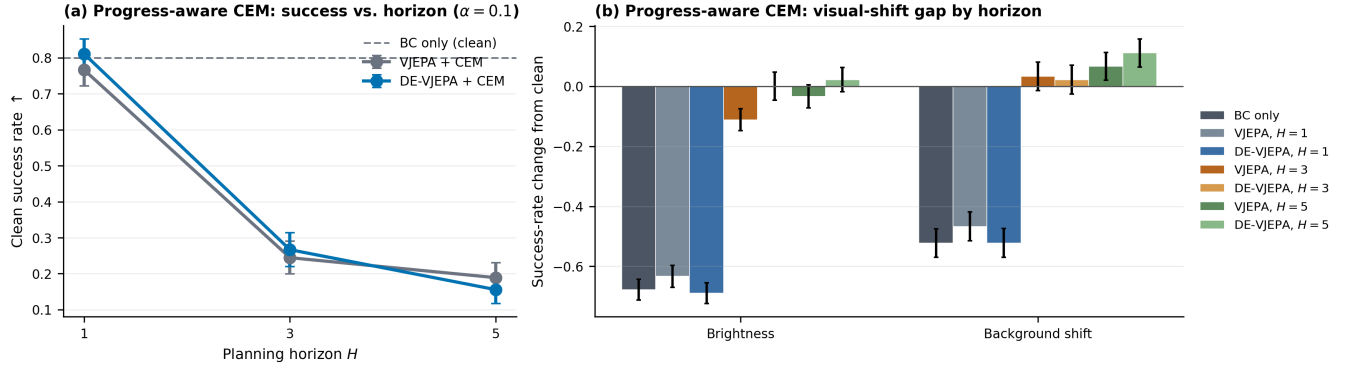


Figure 10: Progress-aware CEM success by planning horizon (left) and visual condition (right). Horizon three yields the largest directional DE-VJEPA advantage, while longer planning does not improve aggregate success.